

HR Attrition Analysis Project Report

1. Introduction:

Employee attrition is a major challenge for organizations as it affects productivity, increases recruitment costs, and impacts overall business performance. Analyzing employee attrition data helps Human Resource (HR) teams identify the key factors influencing employee turnover and take corrective actions.

This project focuses on analyzing HR attrition data using **Power BI** to identify trends, patterns, and insights that can help reduce attrition and improve employee retention.

2. Abstract:

The goal of this project is to perform data cleaning, transformation, and visualization on the HR dataset to uncover insights about employee attrition.

- The dataset consists of **1,470 employee records** with attributes such as age, gender, department, job role, overtime, and education.
- Using **Power Query** and **DAX**, data was transformed into a clean format and KPIs like **Total Employees, Attrition Count, and Attrition Rate** were calculated.
- An interactive **3-page dashboard** was developed in **Power BI** to analyze attrition by various dimensions (age, department, job role, education, overtime).
- The results highlight key drivers of attrition, such as job roles, overtime, and age groups, enabling HR managers to design better retention strategies.

3. Tools Used:

- Excel → For initial dataset storage and exploration
- SQL (PostgreSQL/MySQL) → For querying and analyzing the dataset
- Python (Jupyter Notebook) → For data preprocessing, exploratory data analysis (EDA), and generating insights
- Power Query (Power BI) → For additional data cleaning and transformation
- DAX (Data Analysis Expressions) → For KPI and measure calculations
- Power BI Desktop → For dashboard creation and visualization.

4. Steps Involved in Building the Project:

1. Data Collection: Imported HR attrition dataset into Excel.
2. SQL Analysis: Queried and analyzed the dataset using SQL to understand key metrics such as attrition by department, job role, and gender.
3. Python Analysis: Used Jupyter Notebook for data preprocessing, handling missing values, and exploratory data analysis (EDA).
4. Data Cleaning in Power Query: Further refined and structured the dataset for Power BI.
5. DAX Calculations: Created measures:

- **Attrition Count**
- **Attrition Rate**
- **Highest Attrition by Department**
- **Attrition by Age Band**

6. Dashboard Creation: Designed a 3-page interactive Power BI dashboard

- **Page 1:** Overview (KPIs, department attrition, experience trends).
- **Page 2:** Detailed analysis (age, gender, job role, overtime).
- **Page 3:** Advanced insights (education, attrition trends, attrition rate by role).

5. Conclusion:

The HR Attrition Analysis project successfully demonstrates the use of SQL, Python, and Power BI for data-driven HR analytics. The insights derived from the project highlight critical factors influencing attrition, such as age, job role, and department. By leveraging these insights, organizations can implement targeted strategies to improve employee retention and enhance workforce stability. This end-to-end project showcases a practical application of data analysis in solving real-world HR challenges.

The analysis revealed several key insights:

- **Highest attrition** occurs in **Sales and Research & Development departments**.
- **Younger employees (0–2 years of experience)** are more likely to leave.
- **Overtime** has a strong correlation with employee attrition.
- Certain job roles such as **Sales Representatives and Laboratory Technicians** show higher attrition rates.

These insights can help HR departments implement **targeted retention strategies** such as employee engagement programs, work-life balance improvements, and better career development opportunities.